Cancer treatment failure is often only obvious in hindsight. By the time resistance becomes clinically apparent, the biological processes driving it may have been developing for months or years. Modern oncology generates vast amounts of genomic, transcriptomic, histopathological, and clinical data, yet much of it is still analysed in isolation. At the same time, recent advances in representation learning and foundation models have shown that training on large, heterogeneous datasets can uncover patterns that transfer across tasks and domains. Applying these ideas to cancer research raises an interesting question: can we learn shared representations of tumour biology that generalise across cancer types and help explain why therapies succeed, fail, or stop working?

I have just completed my Master's in Data Science at Uppsala University, where I have gradually focused my interests on healthcare and machine learning. What initially drew me to the field was how data-driven systems can improve healthcare while remaining trustworthy and useful in practice. Early on, I wrote about bias in medical AI systems and how existing limitations in healthcare research risk being amplified as new, powerful models are deployed. In healthcare, predictive performance alone is rarely sufficient. Understanding what a model has learned, whether it is capturing meaningful biological signals or exploiting shortcuts in the data, is equally important.

Through a health innovation course, I studied the European health technology ecosystem and the challenges involved in bringing new technologies into clinical use. In another project, I investigated healthcare data interoperability challenges in Sweden, examining how information moves between systems and organisations. Very recently, I started a pilot collaboration with the Haematology and Oncology Department at King Faisal Specialist Hospital in Saudi Arabia, exploring AI-assisted interpretation of electrophoresis results. While these experiences differ in scope and technical depth, they all reinforced the same observation: healthcare data is rich, fragmented, longitudinal, and captures different parts of the same biological processes.

My strongest technical experience comes from machine learning and software development. My master's thesis focuses on multimodal machine learning for harmful content detection, combining text, image, and contextual information within a unified modelling framework. The work required developing custom PyTorch architectures, designing representation learning approaches, handling large datasets, and making modelling decisions under significant ambiguity. Different domain, but key skills are transferable: integrating heterogeneous data sources, learning useful representations from incomplete information, and identifying which information is genuinely predictive versus merely correlated.

The work on therapy resistance, metastasis, and foundation models all address the challenge of learning representations that capture meaningful biological structure across datasets, cancer types, and prediction tasks. The proposed foundation model component is especially interesting because it reflects a broader shift occurring across machine learning and computational biology. I used to keep up with Med-PaLM and Google’s later works on foundational medical models. Applying that idea across genomic, transcriptomic, methylation, and histopathology data presents substantial technical challenges, but also the possibility of generating insights that

would be difficult to obtain from isolated datasets or task-specific models. It feels ambitious to me, but at the same time, with all the advancements in AI, the right data and support at KI, it seems feasible and would be a fantastic project to work on.

I would be excited to contribute my background in machine learning, multimodal modelling, and healthcare-focused data science to this project while developing the quantitative and biological expertise required to investigate these questions at a deeper level.